Road Risk Analysis Platform

Functional Review and Feedback Report

This document presents the observations, recommendations, and functional expectations identified during the review of the Road Risk Analysis Platform. The objective of this review is to assess the current functionalities, identify opportunities for improvement, and outline additional capabilities that would enhance the platform's effectiveness, usability, and scalability.

Module	Feedback	Expected Improvements
Dashboard	The user interface is not intuitive, and the font family used throughout the application does not provide a modern or professional appearance.	- The overall dashboard design should be improved to enhance usability and user experience.
Path config	The whole functionality does not align with the work flow of the project. We shouldn't be defining paths on which the detections will be done or assigned.	- Remove the Path Configuration module, as the project should not rely on predefined static routes. - Obtain the location of each detection directly from the geolocation embedded in the analyzed video. - Use **Reverse Geocoding** to automatically determine the corresponding road, locality, or administrative zone. - Allow each detected cluster to identify its own location automatically instead of relying on manually assigned paths.
Cluster Config	Defining a cluster radius for each national highway makes the platform depend on static configurations, reducing its flexibility.	- Create a global **Cluster Configuration** section within **Settings**.

		- Allow separate clustering parameters for **Urban (In Agglomeration)** and **Rural (Out of Agglomeration)** areas. - Configure clustering based on the environment rather than individual national highways.
Access Right	- The current access rights configuration does not distinguish between mapping providers or service levels.	- Add a configuration option allowing administrators to select either **OpenStreetMap** or **Google Maps**. - Support both **Free** and **Premium** application modes depending on the selected mapping provider.
Live Map	- On the live map, the routes appear to be drawn as straight lines (plane view) rather than following the actual road network, including road curves and junctions. The routes should ideally follow the real road geometry. - Additionally, the map still assumes predefined paths instead of displaying real detection locations.	- In this module, it will be preferable that since we are not more defining fixed paths, as we land in the page, we have all the detection points clustered over the map pointing exactly where they were captured. - Adding to this, having the possibility to define a trajectory that is a start point and a destination which will trace the path linking the two points and then bring out all the detected clusters in that trajectory. So we can have the setting destination by national highway name , by town name etc …
Hazard Track Page	- Road hazards should be detected and mapped automatically. Manual path creation should not be required. The platform should display detected hazards grouped into geographic clusters for improved visualization.	- Detect and group hazards automatically into geographical clusters. - Improve the visual hierarchy between the cluster list and the detections belonging to each cluster.

	- The **Hazard Track** page is not very intuitive. It is difficult to distinguish between the cluster list and the detection list belonging to a selected cluster. I believe the hierarchy and visual separation between these two elements could be improved. - The appearance of the page is too dull	- Modernize the overall page design to improve readability and user experience. - Replace raw GPS coordinates with human-readable locations obtained through **Reverse Geocoding** (town, district, street, etc.).
Jod	- The **Jobs** page is intended to manage video processing tasks rather than validation activities. Therefore, displaying the **Confirmed** and **Resolved** columns is unnecessary. Likewise, the **Job Details** modal should not include action buttons such as **Validate**, **Confirm**, or **Resolve**. Additionally, the **Upload Video** button does not belong on this page because video upload is a separate input process.	• Remove the **Confirmed** and **Resolved** columns from the Jobs page. • Remove the **Validate**, **Confirm**, and **Resolve** buttons from the Job Details modal. • Move the **Upload Video** functionality to a dedicated menu item within the application navigation. • Keep the Jobs page focused solely on monitoring and managing processing jobs.
Upload Video	- The upload process currently requires users to manually enter metadata such as **Registration Number**, **Device ID**, **Vehicle ID**, and **Vehicle Name**. These details should instead be extracted automatically from the uploaded video whenever available. - Furthermore, users are currently required to assign a path during upload, which is no longer necessary	- Remove all manual vehicle-related input fields and automatically extract the available metadata from the video. - Eliminate the **Path Assignment** field during upload, as detection locations will be determined from extracted geolocation data. - Support uploading multiple videos simultaneously.

	since detections will rely on extracted geolocation information. - The system also limits users to uploading only one video at a time. Additionally, the concept of **Video Source** should represent the storage location from which the video is retrieved (e.g., local storage or cloud storage) rather than the vehicle itself. - Finally, users have no visibility into which AI model will process the uploaded video such as (Pothole, Traffic sign, …).	- Redefine **Video Source** to represent the storage source (e.g., Local Storage, Cloud Drive, Shared Repository) rather than vehicle information. - Add support for uploading videos directly from cloud storage platforms (e.g., Google Drive, OneDrive, shared repositories), allowing the server to retrieve videos automatically for AI processing. - Display the AI model that will be used to process each uploaded video. - Implement a processing queue where multiple uploaded videos are queued automatically. As processing slots become available, queued videos should start execution as individual jobs, allowing the server to process multiple jobs concurrently based on available resources.
Validations	- No need displaying " GPS " column since we already have the location - On which principle are the flags derived ? - When we click on the button action "Cross", we have a popup that says that the detection have been deleted, no detection should be completely deleted for a first move.	- Remove the **GPS** column and display only the resolved textual location obtained through Reverse Geocoding. - Clearly define and document how detection **Flags** are generated and what each flag represents. - Modify the rejection workflow so that rejected detections are **not deleted**. - Allow reviewers to correct incorrectly assigned flags before rejecting a detection. - Store rejected detections in a dedicated training dataset to support continuous retraining and improvement of the AI detection and classification models.

		- Prevent detections or clusters from being created when no valid geolocation can be determined automatically. - When geolocation extraction fails, display the extracted information together with approximately **5 seconds before and 5 seconds after** the detected event, enabling reviewers to identify and manually update the correct location. - Once the location has been corrected, automatically apply **Reverse Geocoding** to generate the corresponding textual location and continue with the validation or rejection process.

Additional Functional Requirements

The following functionalities are proposed as new features to further enhance the platform and support future integration with end-user solutions.

1. Community Reporting Module

A new **Community Reporting** module should be introduced to enable **end users** to actively contribute to road safety by reporting road events directly through the platform. Under this approach, end users become an additional source of information alongside the AI-based road hazard detection system.

End users should be able to report events such as:

- Traffic congestion
- Police presence (Mobile Camera, Hidden Camera, Roadside Checkpoint, etc.)
- Road accidents

- Road hazards
- Road closures
- Blocked lanes
- Map issues
- Adverse weather conditions
- Roadside assistance requests
- Places of interest

Depending on the selected report type, users should be able to provide one or more of the following:

- GPS location (captured automatically)
- Photographs
- Voice recordings
- Text descriptions

All submitted reports should be transmitted to the backend and displayed in the administration dashboard for monitoring and management purposes.

Unlike AI-generated detections, the validity of user reports should not rely solely on administrators. Instead, the platform should implement a **community-based voting system**, allowing end users to collectively validate or invalidate reports over time.

❖ Report Visibility Management

The platform should also support the configuration of **report visibility radii** based on the operating environment.

Two independent configurations should be available:

- **In Town (Urban Areas)**
- **Out of Town (Rural Areas)**

When an end user is located within an urban area, only reports that fall within the configured urban visibility radius should be displayed. Likewise, when the user is outside an urban area, only reports located within the configured rural visibility radius should be visible.

This approach ensures that users receive only relevant reports based on their current environment while avoiding unnecessary information overload.

All visibility radii must be **fully configurable** through the platform settings and should not rely on hard-coded values.

This approach will improve data reliability while ensuring that outdated reports are automatically removed through community feedback.

❖ Backend API Requirements

To support integration between the backend platform and end-user interfaces, the following REST APIs should be made available.

● User Report APIs

Create a New Report

Creates a new road report.

The API should accept a **multipart/form-data** request containing:

- Report type (Accident, Hazard, Traffic, Police, etc.)
- GPS coordinates
- Optional description
- Optional image attachment
- Optional audio attachment

- **Retrieve Nearby Reports**

Returns all active reports located near the user's current position.

- **Request Parameters**

- Latitude
- Longitude
- Search radius

Note: The search radius is **fully configurable** through the platform settings and may vary depending on whether the user is located in an urban or rural environment.

Response

- List of active reports located within the configured search radius.

❖ Retrieve Reports Along a Planned Route

Returns all active reports located within a corridor surrounding a planned route.

Request Parameters

- List of route waypoints
- Corridor width

Note: The corridor width is **fully configurable** through the platform settings and should not rely on a fixed value.

This endpoint enables end users to receive notifications about reported events located ahead on their planned route.

❖ Confirm a Report

Allows users to confirm that a reported event is still present.

Each confirmation increases the confidence score of the report and extends its validity period.

The system should also verify the user's location before allowing a confirmation. A user should only be able to confirm a report if they are located within the configured validation radius of that report. This validation radius should be configured separately for **Urban (In Town)** and **Rural (Out of Town)** environments, ensuring that confirmations originate only from users who are genuinely close to the reported event.

❖ Reject / Expire a Report

Allows users to indicate that a reported event is no longer present.

This community voting mechanism is used to automatically remove outdated reports.

As with confirmations, report rejection should only be permitted when the user is located within the configured validation radius corresponding to the user's environment (Urban or Rural).

❖ AI Road Detection APIs

These APIs expose hazards detected and validated by the AI models.

Retrieve Nearby AI Detections

Returns AI-validated road hazards (such as potholes, speed bumps, road signs, etc.) located near the user's current position.

Request Parameters

- Latitude
- Longitude
- Search radius

Note: The search radius is **fully configurable** through the platform settings and may have different values for urban and rural environments.

The backend performs all spatial filtering before returning the results.

❖ Retrieve AI Detections Along a Planned Route

Returns all validated AI detections located within a corridor surrounding a planned route.

Request Parameters

- Route waypoints
- Corridor width

Note: The corridor width is **fully configurable** through the platform settings, allowing it to be adapted to different operational requirements.

This endpoint enables end users to receive notifications about AI-detected road hazards located ahead on their planned route.

❖ Configuration Requirements

To ensure flexibility across different deployment environments, all spatial parameters used by the platform should be configurable through the **Settings** module rather than being hard-coded.

These parameters include:

- Report visibility radius
- Report validation radius
- AI detection search radius
- Planned route corridor width
- Cluster radius for **Urban (In Town)** areas
- Cluster radius for **Rural (Out of Town)** areas

This configuration allows administrators to optimize the platform according to local road networks, traffic density, and operational requirements while ensuring consistent behavior across all interfaces used by end users.

Based on your report, the **Community Reporting Module** is already defined as the **first additional functionality**.

2. Driver Behaviour Analytics Module

In addition to AI-based road hazard detection and community reporting, the platform should provide the capability to analyze driving behaviour using telemetry data collected from connected vehicles or onboard devices.

The objective of this module is to identify hazardous driving patterns and generate valuable insights that can be used to improve road safety.

The platform should be capable of collecting and analyzing events such as:

- Speeding events (speed limit violations)
- Harsh braking events
- Sudden acceleration events
- Harsh cornering events
- Rapid deceleration events
- Driver stop duration
- Average travelling speed by road segment
- Traffic flow indicators

Each event should be geolocated and stored within the platform to enable spatial analysis.

❖ Risk Zone Identification

By aggregating these driving behaviour events, the platform should automatically identify **high-risk zones**.

For example:

- Areas where a significant number of vehicles repeatedly perform harsh braking may indicate hidden road hazards, poor road conditions, dangerous intersections, or unexpected obstacles.
- Areas with frequent sudden accelerations may reveal unsafe overtaking zones or driver behaviour patterns requiring further investigation.
- Locations with repeated speeding events may indicate sections of road requiring enhanced enforcement or additional traffic calming measures.
- Areas exhibiting combinations of speeding, harsh braking, and accidents can be automatically classified as critical road safety hotspots.

These insights should complement AI detections and community reports to provide a more comprehensive understanding of road safety conditions.

❖ Dashboard Integration

The platform should provide dedicated dashboards and analytical maps displaying:

- Speeding hotspots
- Harsh braking hotspots
- Sudden acceleration hotspots
- High-risk driving corridors
- Driver behaviour heatmaps
- Historical trends and statistics
- Risk indicators by road, district, city, or administrative region

These analytics should assist transportation authorities and road safety agencies in identifying hazardous locations and prioritizing corrective actions.

❖ Backend API Requirements

The backend should expose APIs allowing authorized systems to submit and retrieve driving behaviour events.

The APIs should support:

- Submission of speeding events
- Submission of harsh braking events
- Submission of sudden acceleration events
- Retrieval of nearby driving behaviour events
- Retrieval of driving behaviour events along a planned route
- Retrieval of aggregated driving behaviour statistics
- Retrieval of identified high-risk zones and heatmaps

❖ Configurable Thresholds

The thresholds used to classify speeding, harsh braking, sudden acceleration, and other driving behaviour events should be fully configurable through the platform settings.

Administrators should also be able to configure:

- Speed thresholds by road category
- Braking intensity thresholds
- Acceleration thresholds
- Risk score calculation parameters
- Event aggregation radius (Urban and Rural configurations)
- Heatmap generation parameters

This flexibility will allow the platform to adapt to different operational requirements and national road safety standards while continuously improving the identification of high-risk areas.

3. Predictive Cooperative Collision Prevention Module

Overview

In addition to road hazard detection and community reporting, the platform should be extended with a **Predictive Cooperative Collision Prevention Module**. This functionality transforms the Road Risk Analysis Platform from a passive road hazard detection solution into an intelligent road safety ecosystem capable of predicting potential collisions before they occur.

The objective of this module is not to replace existing vehicle tracking systems but rather to provide an additional layer of road intelligence that continuously evaluates the movement of connected vehicles, predicts future trajectories, and proactively alerts drivers whenever a potential collision risk is identified.

Unlike conventional GPS tracking solutions, which simply display the current location of vehicles, this module introduces an Artificial Intelligence prediction engine capable of understanding vehicle behaviour, road geometry, and future vehicle interactions.

System Principle

Each participating vehicle is equipped with the Road Risk Analysis mobile application, which operates continuously in the background while the vehicle is in motion.

The application periodically acquires and transmits operational data, including:

- GPS Position
- Vehicle Speed
- Heading (Direction of Travel)
- Timestamp
- Vehicle Identifier
- Vehicle Type (Bus, Taxi, Truck, Private Vehicle, etc.)

These data packets are transmitted securely to the central Road Risk Analysis Platform using available communication networks (4G, 3G, Wi-Fi or future V2X communication technologies).

The platform continuously maintains a live representation of all active vehicles connected to the system.

Road Intelligence Broker

Rather than allowing every connected vehicle to communicate directly with every other vehicle, the platform introduces a centralized component referred to as the **Road Intelligence Broker**.

The Road Intelligence Broker is responsible for continuously analysing all incoming vehicle information and determining which surrounding vehicles are actually relevant to each connected user.

Instead of simply identifying nearby vehicles based on geographical distance, the broker performs intelligent filtering by considering:

- Current road segment
- Driving direction
- Relative speed
- Predicted trajectory
- Road geometry
- Junctions
- Blind curves
- Elevation
- Historical traffic behaviour
- Configurable safety radius

Only vehicles that may realistically interact with one another within a predefined prediction horizon are considered potential hazards.

This significantly reduces unnecessary data exchange while improving the quality and relevance of driver notifications.

Artificial Intelligence Predictive Engine

The core of this functionality is the **Predictive Artificial Intelligence Engine**.

Rather than reacting after dangerous situations occur, the AI continuously predicts the future position of every connected vehicle several seconds ahead.

Using real-time telemetry together with digital road network information, the prediction engine estimates:

- Future vehicle trajectories
- Estimated arrival time at intersections
- Relative vehicle speed
- Lane convergence
- Potential crossing points
- Vehicle encounter probability
- Time-To-Collision (TTC)
- Collision probability score

The AI model continuously recalculates these predictions whenever new vehicle information is received, allowing the system to adapt instantly to speed changes, braking events, route deviations, or unexpected driver behaviour.

The prediction process enables the platform to identify dangerous situations well before they become visible to drivers.

Risk Evaluation

For every predicted vehicle interaction, the platform computes a dynamic risk score using several contextual parameters, including:

- Distance between vehicles
- Relative speed
- Direction of travel
- Predicted point of convergence
- Time-To-Collision
- Road curvature
- Road elevation
- Visibility constraints
- Historical accident data

- Weather conditions (future integration)

The calculated score is then classified into configurable alert levels.

Example:

- Low Risk
- Moderate Risk
- High Risk
- Critical Collision Risk

Only situations exceeding predefined safety thresholds generate driver notifications.

Driver Notification Mechanism

Whenever a significant collision probability is detected, the platform immediately delivers an alert to the affected vehicles.

Depending on the calculated risk level, notifications may include:

- Voice guidance
- Audible alarms
- Screen notifications
- Device vibration
- Navigation overlay warnings

Example notifications include:

"Warning. Vehicle approaching from the opposite direction."

"Reduce speed. Dangerous blind curve ahead."

"Collision risk detected. Another vehicle is approaching the same junction."

The notification system is designed to minimize driver distraction while maximizing reaction time.

Road Context Awareness

Unlike traditional fleet management platforms, collision prediction is performed within the context of the actual road network.

The platform uses digital mapping services (OpenStreetMap, Google Maps or equivalent providers) to associate every vehicle with its corresponding road segment.

This enables the prediction engine to distinguish between situations such as:

- Vehicles travelling in opposite directions on the same road
- Vehicles approaching hidden junctions
- Blind curves
- Mountain passes
- Road intersections
- Highway merging lanes
- Multi-lane carriageways

Consequently, two vehicles separated by only a few metres but travelling on independent roads will not generate unnecessary alerts.

End-User Application

A dedicated Road Risk Analysis mobile application should be provided for end users.

The application becomes the operational interface between drivers and the central Road Risk Analysis Platform.

Besides supporting the Community Reporting functionality described previously, the application should provide:

- Live navigation
- Nearby AI-detected hazards
- Community reported events
- Predictive collision warnings

- Safe route recommendations
- Intelligent voice notifications
- Planned route monitoring
- Offline map support
- Automatic synchronization when network connectivity is restored

The application therefore serves as both an information provider and an intelligent safety assistant throughout the journey.

Communication Architecture

The communication model follows a cooperative architecture.

1. Each connected vehicle periodically transmits its position and movement information to the Road Risk Analysis Platform.
2. The Road Intelligence Broker receives all telemetry data.
3. The Artificial Intelligence Prediction Engine continuously predicts future vehicle trajectories.
4. The Risk Assessment Engine evaluates possible collision scenarios.
5. Only the vehicles concerned by a predicted risk receive an alert.
6. Drivers take the appropriate preventive action.

This architecture minimizes network traffic while ensuring that only relevant safety information is exchanged.